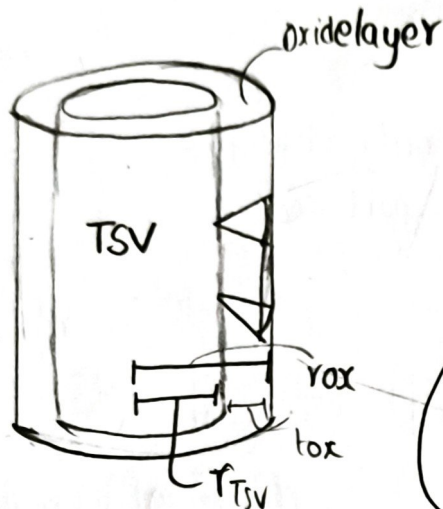
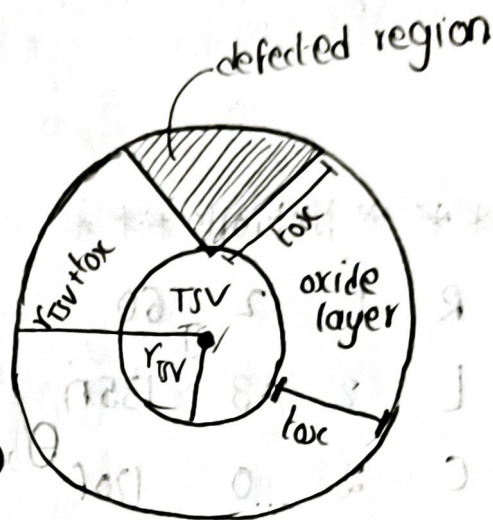
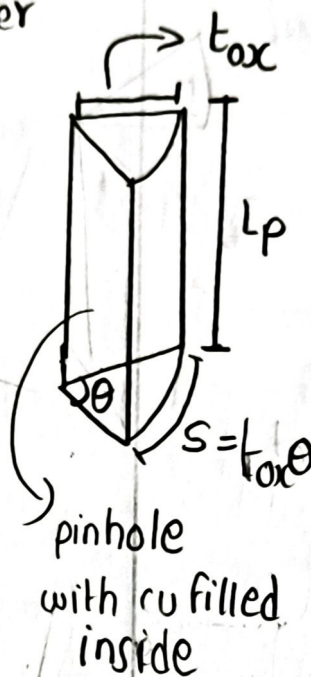


10/06/26 DAY 15 "Kamineni Dhana Sri Koorthi"

▲ Correction of R_{sht} & L_{sht}



$$r_{ox} = r_{TSV} + t_{ox}$$



$$R_{sht} = \frac{\rho_T t_{ox}}{L_p [t_{ox} \theta]}$$

$$= \frac{\rho_T}{L_p \theta}$$



ρ_T = resistivity of copper

$\frac{\rho l}{A}$
current path.
A = area let to current flow.

$$R_{TSV} = \frac{\rho_{TSV} \frac{l_{travel}}{length}}{\pi (r_{TSV})^2}$$

→ per surface area to I flow
similar to this

$$L_{sht} = \frac{\mu_0 t_{ox}}{2\pi} \left[\ln \left[\frac{2t_{ox}}{r_{eq}} - 1 \right] \right]$$

from $\pi r^2 = A$

$$r_{eq} = \sqrt{\frac{A}{\pi}}$$

$$A = L_p t_{ox} \theta$$

= self inductance of straight wire cylinder

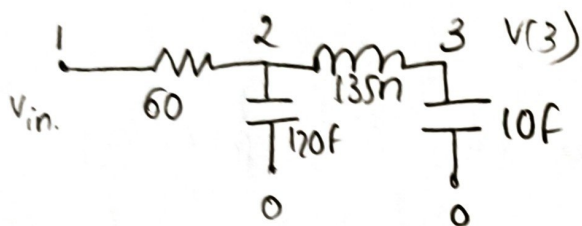
$$L = \frac{\mu_0 l}{2\pi} \left[\ln \left[\frac{2l}{r} \right] - 1 \right]$$

l = current path.

r = effective conductor radius.

If defect is not there then r is conductor or cylinder radius.

T-Network.

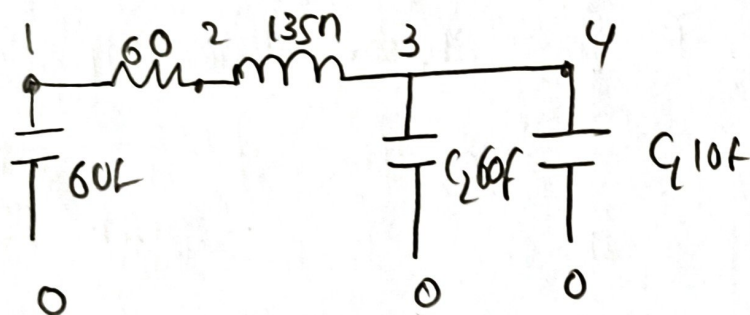


only changing
part code.

*** Network ***

R	1	2	60
L	2	3	135n
C	2	0	120f
C1	3	0	10f

Pi-Network.



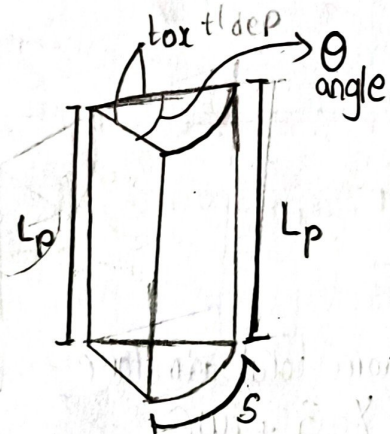
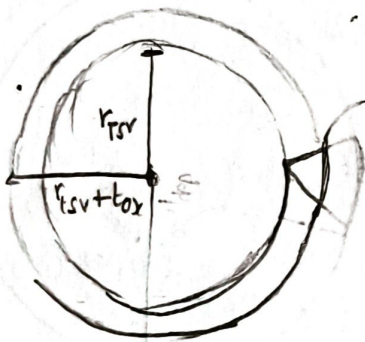
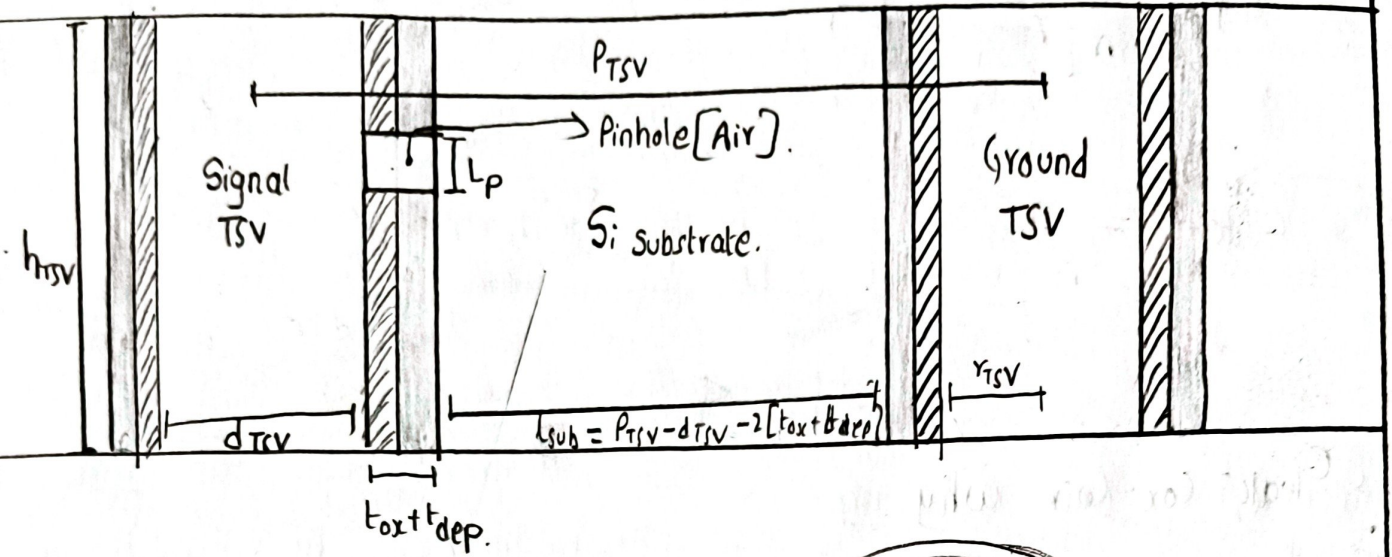
*** Network ***

R	1	2	60
L	2	3	135n
C ₁	1	0	60f
C ₂	3	0	60f
C _L	4	0	10f.

Basic flow:
workspace o/
↓
applications
↓
accessories
↓
Text editor
(code writing)
↓
save it in files with
[folder] .sp extension
↓
Go to that folder
↓
open in terminal
(right click)
↓
Type hspice
↓
long info +
Enter input file:
(whatever you save that
name.sp)
↓
press enter
↓
Enter o/p file name:
[name.lis]
press enter
↓
Hspice extension
press enter.

↓
in info if the code is
correct it will
show
→ hspice job concluded.
↓
if not it will show error
↓
Files get generated.
(ho, sto, ico. →)
↓
type wv
↓
In interface click open
waveform
↓
select .tro file for
wave form
↓
select the top level
file at left side
↓
select v(1) or v(3)
to get wave form
display.

DEFECTED TSV [C_{leak} & G_{leak}]



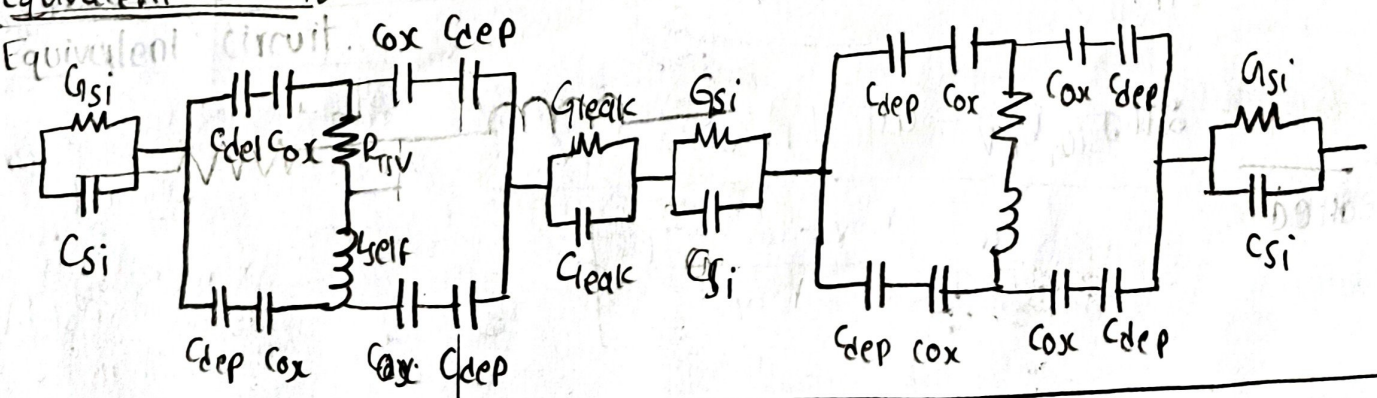
$s = \text{arclength}$ & $\theta = \text{radians}$

$$s = r\theta \quad s = [t_{ox} + t_{dep}] \theta \text{ [m]}$$

$$s = [t_{ox} + t_{dep}] \theta \text{ [m]}$$

Equivalent circuit-

Equivalent circuit. C_{ox} C_{dep}



Equations.

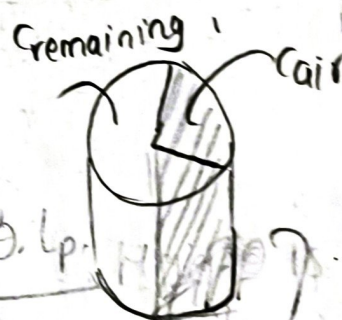
$$C_{air} = \frac{\epsilon_0 \epsilon_{air} \theta L_p}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]}$$

$$C_{leak} = C_{ox} - C_{air}$$

$$C_{leak} = \left[\pi \epsilon_0 \epsilon_{ox} \left[\frac{h_{TSV}}{2} \right] - \frac{\epsilon_0 \epsilon_{air} \theta L_p}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \right]$$

$$C_{leak} = C_{ox} - C_{air} \text{ why?}$$

↓
how much capacitance we lost because of this air defect.



$$C_{ins} = \frac{C_{dep} C_{ox}}{C_{dep} + C_{ox}}$$

$$G_{leak} = G_{Si} - Y \text{ [admittive capacitance]}$$

since $G_{air} \approx 0$

$$Y = j\omega C$$

∴ from total admittance
 $Y = G_{air} + j\omega C$

$$J = \frac{\theta L_p}{2\pi Y_{TSV}}$$

$$G_{leak} = G_{Si} - Y_c$$

$$G_{leak} = \frac{2\pi \sigma_{Si} L_{TSV}}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} - 4\pi^2 J f \epsilon_0 \epsilon_{air} r_{TSV}$$

edited

value Table.

Symbol	value
J	40% caluabk
θ	$\pi/2$
L_p	4um

data taken from a novel TSV made with fault

modelling & simulation of TSV by Chao liu